Evidence-Gated Tool Use for Reliable CAR-bench Agents

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Abstract

Our Track 1 agent obtains 63.3% official Pass^3 and 87.3% Pass@3 on the full 125-task CAR-bench public test set (375 episodes). Its split Pass^3 scores are 74% Base, 52% Hallucination, and 64% Disambiguation. The agent uses Claude Sonnet 4.6 [Anthropic, 2026], but it does not delegate reliability to the model. A deterministic Decision Manager checks visible capabilities, evidence, ambiguity, policy constraints, and completion claims before it emits an A2A response. It invokes bounded candidate sampling only on high-risk turns. On a fixed 18-task development slice, this design improves Pass^3 from 55.6% for a prompt-only agent to 94.4%. The public test also defines the remaining work: 29 tasks pass in only one or two trials, and missing tool response fields obtain only 18.2% Pass^3. These results show that deterministic gates reduce systematic errors, while grounded selection is still necessary to convert latent capability into consistent success.

1 Result and Motivation

CAR-bench evaluates tool-using agents under normal operation, missing capabilities, and ambiguous requests [Kirmayr et al., 2026]. A task contributes to Pass^3 only when all three independent trials pass. Pass@3 instead records whether at least one trial passes [Yao et al., 2024]. The gap between these metrics separates latent capability from reliable behavior.

Our submission uses the frozen image digest sha256:9c69fc4a.... On the public test set, it scores 285/375 episodes and 63.3% official Pass^3. The official value is the unweighted mean over the three task families. The 24.0-point gap to Pass@3 shows that the agent often knows a valid path but does not select it consistently.

The design follows a simple rule: the language model proposes an action, but visible evidence authorizes it. This rule connects classical preconditions [Fikes and Nilsson, 1971], affordance gating [Ahn et al., 2022], and belief-state control under partial observation [Kaelbling et al., 1998]. It also avoids a known weakness of prompt-only abstention: a model can repeat a rule without applying it at the point of commitment.

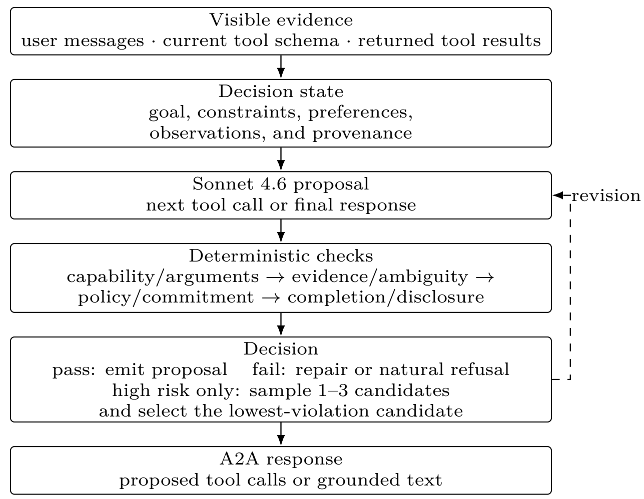

Figure 1: Track 1 assistant step. Solid arrows show the normal path. The dashed arrow is a bounded, risk-gated revision path. All checks use A2A-visible evidence.

2 Decision Manager

Figure 1 shows one assistant step. The agent reads only the user messages, the tool schema supplied for the current turn, and returned tool results. It does not inspect evaluator state and does not execute benchmark tools.

Decision state. The state records the active goal, hard constraints, soft preferences, observed values, and the source of each value. A priority label separates required conditions from optional preferences. This prevents an optional condition, such as current plug availability, from blocking an otherwise valid charging stop.

Capability and evidence checks. The first check rejects tool names or arguments that are absent from the current schema. The second check blocks a state change when a required value is unknown. It first uses available information sources, such as preferences or returned state, and asks the user only when the ambiguity remains. These checks are deterministic and add no LLM call. They are related to contract-grounded tool execution [Liu et al., 2026].

Commitment and terminal checks. Before a write, the manager verifies required reads, confirmation, route constraints, and policy order. After a write, it checks that the final text matches observed tool results. It also enforces route-choice, toll, and alternative-route disclosures.

Risk-gated selection. Low-risk proposals use one draft. High-risk proposals can use one to three additional candidates with preference-first, constraint-first, or uncertainty-first framing. The deterministic checks score each candidate. This step targets variance after the gates remove systematic invalid actions.

3 Evaluation Protocol

We developed on train tasks and froze the submitted image before this evaluation. The public test run uses all 125 tasks and three trials per task: 50 Base, 50 Hallucination, and 25 Disambiguation tasks. Thirteen disjoint chunks make the run restartable. Metrics in this report are reconstructed from the 375 final episode records, not averaged over chunks. The selected artifacts contain no missing task-trial pairs, duplicate pairs, empty trajectories, or infrastructure-error episodes.

The development ladder uses one fixed 18-task slice with six tasks from each family. It supports ablation and promotion decisions, but it is not an estimate of public-test performance.

4 Experiment Program

We treated each failure cluster as a falsifiable hypothesis. Tool and policy checks follow operator preconditions and affordance gating [Fikes and Nilsson, 1971; Ahn et al., 2022]. Evidence provenance and ask-versus-act decisions follow belief-state control under partial observation [Kaelbling et al., 1998]. Candidate sampling targets variance only after deterministic checks remove invalid actions [Wang et al., 2023]. This ordering separates systematic bias from stochastic selection error.

Each challenger passed a focused task gate before the fixed 18-task, three-trial development slice. We promoted it only after a score gain with no task-family regression. Broader unseen-train and public-test runs followed freeze. We define descriptive reliability loss as $\mathcal{L}_{\text{rel}} = 1 - \text{Pass}^3$; it is an evaluation summary, not a trained objective.

The prompt-only baseline scored 38/54 and 55.6% Pass^3. E7 risk-gated selection reached 49/54 and 88.9%. A non-factorial E9+E10 bundle improved Hallucination from 16/18 to 18/18 but fell to 48/54 overall; we rejected it and consolidated its useful mechanisms. Decision Manager v1.1 then reached 53/54 and 94.4% Pass^3. These are cumulative staged comparisons. The prompt baseline used a different serving path, so the full change is not a single-variable ablation.

Public test	Raw	P^1	P^2	P^3	P@3
Base	124/150	84%	74%	74%	90%
Hallucination	102/150	60%	56%	52%	80%
Disambiguation	59/75	84%	68%	64%	92%
Official macro	285/375	76.0%	66.0%	63.3%	87.3%

Table 1: Full public-test results for the frozen Track 1 submission. The official Pass metrics are unweighted means over task families.

Hallucination subtype	Tasks	Raw	0/3	P^3	P@3
Missing tool	29	61/87	5	55.2%	82.8%
Missing parameter	10	27/30	0	80.0%	100.0%
Missing response field	11	14/33	5	18.2%	54.5%

Table 2: Hallucination subtypes. Missing response fields are the main capability-model weakness.

A v1.2 challenger also scored 53/54 but moved the only miss from Disambiguation to Base. We retained v1.1 under the no-regression rule. A later tied challenger lost a matched 15-task unseen-Disambiguation duel: 34/45 raw and 60.0% Pass^3 versus 36/45 and 66.7% for v1.1. An early Opus run scored 40/54 and is only contextual because both model and architecture stage differ.

5 Results

Table 1 gives the main result. Base is the strongest family. Hallucination has the lowest Pass^3 and the largest latent-capability gap. Disambiguation has 64% Pass^3 but 92% Pass@3, which shows that most tasks are solvable but not stable.

Table 2 gives the most important failure result. The manager handles removed parameters well, but it often treats an existing tool as sufficient even when a required result field is absent. A schema-aware observation contract must represent required output fields, not only tool and argument availability.

6 Failure Analysis and Limits

The public test contains 79 reliable 3/3 tasks, 29 tasks with one or two successful trials, and 17 systematic 0/3 tasks. Thus, the largest reachable gain is not a larger model alone. The agent needs better grounded selection on the 29 unstable tasks and new mechanisms for the 17 systematic failures.

The shared systematic classes are partial result schemas, occupied-seat climate bundles, route replacement with charging follow-up, and internal preference binding. Action and required-tool coverage failures also show that the manager can authorize a valid local step without completing the full user goal. A typed subgoal ledger is the next general mechanism.

The 94.4% development Pass^3 falls to 63.3% on public test. This is a 31.1-point generalization gap across different task sets. We therefore treat the public result, not

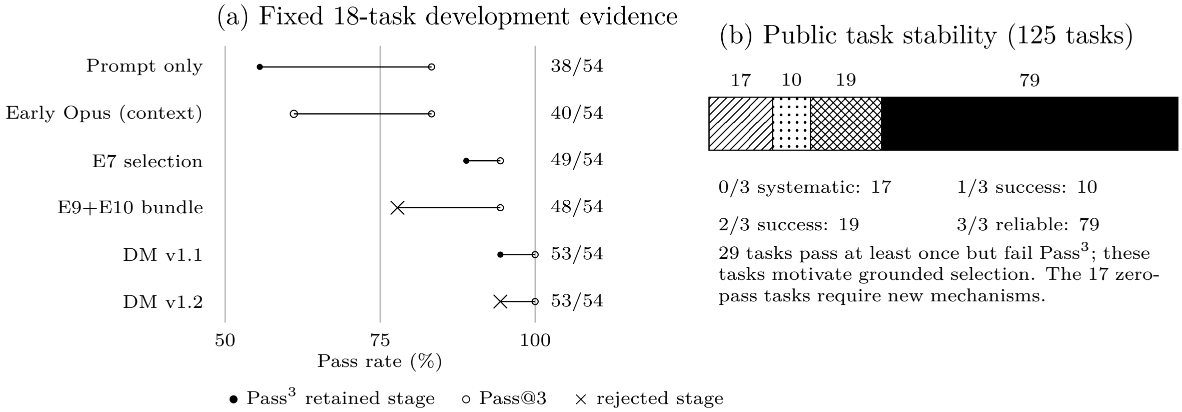

Figure 2: Track 1 experiment trajectory. Horizontal segments show the Pass^3–Pass@3 consistency gap. Rows are independent staged runs, not a smooth training curve. On the retained path, $\mathcal{L}_{\text{rel}}$ falls from 44.4% to 11.1% to 5.6% on the fixed development slice.

the development peak, as the estimate of current reliability. Hidden-test results and ranking are not included.

7 Conclusion

The submitted Track 1 agent combines a capable model with deterministic authorization. This design obtains 63.3% official Pass^3 on the full public test and exceeds its prompt-only development baseline by 38.8 points. The remaining gap has two parts: unstable selection on tasks that sometimes pass, and missing observation contracts on tasks that never pass. Future work should address these parts separately.

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